Problem 1: PPN and Gravitational Red Shift

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Problem

Question: Is the experiment of Vessot Levine sensitive enough to say anything about the parameters β and γ? Is the third-order doppler effect important in analyzing the experiment?[1]

This problem is an exercise in applying the Parameterized post-Newtonian framework to particular prediction. Specifically, by solving this problem, we can deduce whether the experimental results obtained by an experiment such as Vessot and Levine (1980) allows one to determine PPN parameters. Before presenting a solution to the problem, I will briefly sketch both the frequency shift predicted by the equivalence principle; the Vessot and Levine experiment; and the PPN framework.

The gravitational frequency shift is a consequence of the Equivalence Principle. If it is accepted that uniformly frame is entirely equivalent to a uniform (or local) gravitational field, then it must be the case that the rates of emission and reception for spatially separated observers are different. That is to say that frequencies are shifted, preportional to the gravitational field strength. All theories of gravity that are reconciled with the Equivalence Principle—which is to say, all modern theories of gravity—will predict a gravitational frequency shift. The equivalence principle explicitly predicts that for a detector B and an emitter A the rates of emission and reception (i.e. frequencies ω_B and ω_A) are related by the difference in the gravitational field strengths[2]

$$\omega_B = \left(1 + \frac{\Phi_A - \Phi_B}{c^2}\right)\omega_A. \tag{1}$$

[1] Question 10.2 Hartle
[2] See Hartle, 113-119

As we will see shortly, the Vessot and Levin experiment involves the frequency shift involving signals on the surface of the Earth and in orbit. So we can apply (1) to a situation where a signal is emitted from the surface of the Earth and measured by an observer far away. In this case the gravitational potential on the surface is

$$\Phi = -\frac{GM_\oplus}{R_\oplus} \tag{2}$$

where $M_\oplus$ and $R_\oplus$ is the mass and radius of the Earth respectively. The potential as $r \to \infty$ is zero. Thus we have

$$\omega_\infty = \left(1 - \frac{GM_\oplus}{R_\oplus c^2}\right)\omega_\oplus. \tag{3}$$

This problem can treated in General Relativity by approximating the gravitation field using the Schwarzchild geometry, produced by a spherically symmetric body. The Schwarzchild Metric, for the spacetime approximation, is given in geometrical units

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2). \tag{4}$$

The gravitational redshift, (3), can be recovered in a clever way. First we note that the Killing vector has temporal and spherical symmetry; the metric (4) is independent of the coordinates t and ϕ. For our purposes we will exploit the temporal symmetry and define a Killing vector

$$\xi^\alpha = (1, 0, 0, 0). \tag{5}$$

We then define the conserved scalar product of the four-velocity and this Killing vector along a geodesic

$$\boldsymbol{\xi} \cdot \boldsymbol{u} = \text{const.} \tag{6}$$

Next we set up on problem as following: an observer travels along a world line with four-velocity $\boldsymbol{u}_{obs}$ measures the frequency of light by measuring its energy. The energy an observer measures is related to the four-momentum of the light and their own four-velocity by

$$E = -\boldsymbol{p} \cdot \boldsymbol{u}_{obs}. \tag{7}$$

Furthermore we note that we can relate the energy of a photon to its frequency using the Planck relation. This results in

$$\hbar\omega = -\boldsymbol{p} \cdot \boldsymbol{u}_{obs}. \tag{8}$$

We note that the spatial components of the observers four-velocity are all zero. Using the normalization condition of four-velocity, we have

$$g_{tt}[u^t_{obs}]^2 = -1. \tag{9}$$

Using the temporal metric element from (4) we can specify the four-velocity component

$$u^t_{obs} = \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}}. \tag{10}$$

Using the Killing vector () we can thus write the observers four-velocity as

$$\boldsymbol{u}_{obs} = \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}} \boldsymbol{\xi}. \tag{11}$$

The frequency of light as measured by an observer on the surface of the Earth is

$$\hbar\omega_{R_\oplus} = \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}} (\boldsymbol{p} \cdot \boldsymbol{\xi}) \,|_{r=R_\oplus}. \tag{12}$$

Similarly, the frequency of light as measured by an observer far away as $r \to \infty$ is

$$\hbar\omega_\infty = \left(1 - \frac{2M}{r}\right)^{-\frac{1}{2}} |_{r\to\infty} \, (\boldsymbol{p} \cdot \boldsymbol{\xi}) \,|_{r\to\infty}. \tag{13}$$

The Schwarzchild coefficient reduces to one in the limit where $r \to \infty$. We finally note the conserved scalar product as defined by (). This permits () to be substituted for () and we find

$$\frac{\omega_\infty}{\omega_{R_\oplus}} = \left(1 - \frac{2M}{r}\right)^{\frac{1}{2}}. \tag{14}$$

This result can be expanded in powers of $2M/R$ when that is small. The first order term recovers the redshift as prediced solely by the Equivalence

Principle in ().

The predictions of the gravitational frequency shift have been tested experimentally to quite amazing degrees of accuracy. It is worth noting that these experiments do not stricly test General Relativity. In fact, that is the entire point of this problem. All theories of gravity, that are in agreement with the Equivalence Principle, predict the frequency shift. Thus, experiments such as those described below are really tests of the Equivalence Principle; they do not, in and of themselves, discriminate between different forms of metric theories of gravity.

Vessot and Levin (1980)[3] was an experiment conducted to test the relativistic frequency shift as predicted by the Equivalence Principle. The experimental design is quite ingenious in the fact that it was able to eliminate a major source of error. There are two sources of relativistic frequency shift: the doppler shift, a consequence of the relative motion of the spacecraft and the ground system, and the gravitational redshift. The former can be expressed as a Taylor series, whose first and second order terms are of principle importance. The first order term is on the order $\frac{V}{c}$, where V is the relative velocity. In the experiment this would result in a shift on the order 10^{-5}. The second order doppler shift would thus be on the order 10^{-10}. On the other hand the gravitational redshift would be on the order 10^{-10} as well. It is clear that the first order doppler shift would dominate. Any experiment would need to isolate or remove the first order doppler effect in order to probe the gravitational phenomena.

Figure 1 is a sketch of the Doppler cancellation and tracking system as used by Vessot et al. Two masers are used, with two-way maser-transponder-mixer system used to isolate the first order Doppler shift. The authors of the paper note that the first order Doppler effects "were removed by subtracting in mixer M3 one-half the two-way Doppler shift obtained from mixer M2 by comparing the separate two-way transponder links." The result is a measurement of the frequency shift whose effects include only that of second order Doppler effect and the gravitational redshift. The former can be isolated due to the fact that the velocities of the ground system and the probe (spacecraft) system are known.

Parameterized post-Newtonian (PPN) formalism is a framework in which different metric theories of gravity can be compared with each other, and be

[3] R. F. Vessot et al., "Test of relativistic gravitation with a space-borne hydrogen maser," *Physical Review Letters*, vol. 45, no. 26, pp. 2081–2084, Dec. 1980. doi:10.1103/physrevlett.45.2081

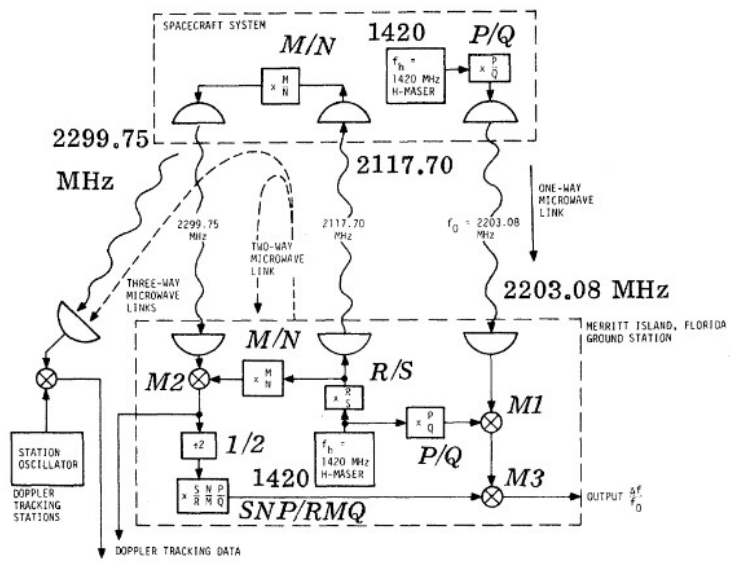

Figure 1: Schematic of the Vessot & Levine gravitational redshift experiment.

experimentally tested. Metric theories of gravity are those theories that posit that gravitation arises from curvature of spacetime. These theories include General Relativity, the most well known and tested, as well as others. Metric theories of gravity, for the most part, provide a spherically symmetric with the form

$$ds^2 = -A(r)(cdt)^2 + B(r)dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2). \tag{15}$$

The functions $A(r)$ and $B(r)$ are specified by the first post-Newtonian corrections that are in agreement with the weak-field limit

$$A(r) = \left(1 - \frac{2GM}{c^2 r} + 2(\beta - \gamma)\left(\frac{GM}{c^2 r}\right)^2\right) \tag{16}$$

$$B(r) = 1 + 2\gamma\left(\frac{GM}{c^2 r}\right) \tag{17}$$

Differing metric theories of gravity will propose different values for the PPN parameters β and γ. General Relativity, for example, has values $\beta = 1$ and $\gamma = 1$. The utility of the PPN framework is material observations, such

as the deflection of light by the Sun, or the precession of the perihelion of a planets orbit, or the Shapiro time delay, can be expressed in terms of the PPN parameters. Thus experiments can be used to determine the values of the PPN parameters; and compare them with the proposed values from a given metric theory of gravity.

Solution

We first need to derive the gravitational red shift in terms of the PPN metric. The line element of the PPN metric has the form

$$ds^2 = -A(r)(cdt)^2 + B(r)dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2). \tag{18}$$

In this case the coefficients (r) and (r) are parameterized by the PPN parameters. To the second-order these have the form

$$A(r) = \left(1 - \frac{2GM}{c^2 r} + 2(\beta - \gamma)\left(\frac{GM}{c^2 r}\right)^2\right) \tag{19}$$

$$B(r) = 1 + 2\gamma\left(\frac{GM}{c^2 r}\right) \tag{20}$$

Now we can setup the general gravitational redshift problem. Recall the structure: we take an emitter and observer at rest. In the case of Vessot Levine, we are considering an emitter on the surface of the Earth, and an observer at rest some large R away. The observer has a four-velocity $\boldsymbol{u}_{obs}$. The energy measured by the observer is

$$E = -\boldsymbol{p} \cdot \boldsymbol{u}_{obs}. \tag{21}$$

We relate this scalar product to the frequency of the photon using Planck's relation

$$\hbar\omega = -\boldsymbol{p} \cdot \boldsymbol{u}_{obs}. \tag{22}$$

The spatial components of the observers four-velocity are zero; that is $u^i_{obs} = 0$. We can then use the normalization constraint on the four-velocity

$$\boldsymbol{u}_{obs} \cdot \boldsymbol{u}_{obs} = g_{\alpha\beta} u^\alpha_{obs} u^\beta_{obs} = -1. \tag{23}$$

With the vanishing spatial components we are left with

$$g_{tt}[u^t_{obs}]^2 = -1. \tag{24}$$

Using (1) we can write

$$-A(r)[u^t_{obs}]^2 = -1. \tag{25}$$

Explicitly then, the only non-vanishing part of the observers four-velocity is

$$u^t_{obs} = A(r)^{-\frac{1}{2}}. \tag{26}$$

We can also use the Killing vector ξ, as described above, to express the temporal component of the four-velocity as

$$u^\alpha_{obs} = A(r)^{-\frac{1}{2}}\xi^\alpha. \tag{27}$$

Explicity the four-velocity is

$$\boldsymbol{u}_{obs} = A(r)^{-\frac{1}{2}}\boldsymbol{\xi}. \tag{28}$$

From this point forward we can exploit the symmetry of the spacetime, exactly as it is done in the Schwarzchild instance. We use (5) and evaluate using $r = R_\oplus$, where $R_\oplus$ is the radius of the Earth,

$$\hbar\omega_{R_\oplus} = A(R)^{-\frac{1}{2}}\left(\boldsymbol{p}\cdot\boldsymbol{\xi}\right)|_{r=R_\oplus}. \tag{29}$$

We also consider the distance observer, where for our purposes we let $r \to \infty$

$$\hbar\omega_\infty = A(\infty)^{-\frac{1}{2}}\left(\boldsymbol{p}\cdot\boldsymbol{\xi}\right)|_{r\to\infty}. \tag{30}$$

$A(\infty)$ reduces to unity which simplifies the expression. The Killing vectors permit us to exploit the conserved scalar product

$$\boldsymbol{p}\cdot\boldsymbol{\xi} = \text{const}, \tag{31}$$

in order to relate the two frequencies. We use (13), simplified with the PPN coefficient reducing to unity, and insert the result into (12). This yields

$$\hbar\omega_{R_\oplus} = A(R_\oplus)^{-\frac{1}{2}}\hbar\omega_\infty, \tag{32}$$

and the gravitational red shift resolves to

$$\frac{\omega_\infty}{\omega_{R_\oplus}} = A(R_\oplus)^{\frac{1}{2}}. \tag{33}$$